Practice Questions for 11L2-svd-pca-recommender

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1 Questions

1. According to the lecture, what is the main motivation behind using matrix factorization?

   A. To increase the dimensionality of the data

   B. To find simpler structures in high-dimensional data

   C. To make the data more complex

   D. To remove all the data

   E. To make the data harder to visualize

2. In the context of matrix factorization for recommender systems, what do the columns of matrix B typically represent?

   A. User preferences

   B. Latent factor values for movies

   C. The original rating matrix

   D. The number of users

   E. The number of movies

3. According to the lecture, what is the primary difference between SVD and PCA?

   A. PCA is a more general operation than SVD

   B. SVD is a more general operation than PCA

   C. PCA provides more information than SVD

   D. SVD is only used for dimensionality increase

   E. PCA is only used for matrix factorization

2 Answers

1. According to the lecture, what is the main motivation behind using matrix factorization?

   A. Answer A is incorrect because matrix factorization is used to reduce the dimensionality of the data, not increase it.

   B. Answer B is correct because matrix factorization aims to find simpler, underlying structures in high-dimensional data, as stated in the lecture.

   C. Answer C is incorrect because matrix factorization aims to simplify the data, not make it more complex.

   D. Answer D is incorrect because matrix factorization aims to preserve the essential information in the data, not remove it.

   E. Answer E is incorrect because matrix factorization can help with visualization by reducing the dimensionality of the data.

2. In the context of matrix factorization for recommender systems, what do the columns of matrix B typically represent?

   A. Answer A is incorrect because user preferences are represented by the rows of matrix A.

   B. Answer B is correct because the columns of matrix B represent a vector of latent "factor" values for each movie, as explained in the lecture.

   C. Answer C is incorrect because the original rating matrix is represented by S.

   D. Answer D is incorrect because the number of users is represented by the dimension M.

   E. Answer E is incorrect because the number of movies is represented by the dimension N.

3. According to the lecture, what is the primary difference between SVD and PCA?

   A. Answer A is incorrect because SVD is the more general operation.

   B. Answer B is correct because SVD is a general operation defined on all matrices, while PCA is a data analysis technique for dimensionality reduction, which can be achieved by computing the SVD of the matrix.

   C. Answer C is incorrect because SVD provides more information than PCA, specifically the matrix U.

   D. Answer D is incorrect because SVD is used for dimensionality reduction.

   E. Answer E is incorrect because PCA is a dimensionality reduction technique, not just matrix factorization.